

Definizioni per il λ -calcolo (non tipato)

Sintassi	$e ::= x \mid \lambda x.e \mid e\ e$
Variabili libere	$FV(x) = \{x\}$ $FV(e_1\ e_2) = FV(e_1) \cup FV(e_2)$ $FV(\lambda x.e) = FV(e) \setminus \{x\}$
Capture avoiding substitution	$x\{x := e\} = e$ $y\{x := e\} = y$ se $x \neq y$ $e_1e_2\{x := e\} = (e_1\{x := e\})(e_2\{x := e\})$ $(\lambda y.e_1)\{x := e\} = \lambda y.(e_1\{x := e\})$ se $y \neq x$ e $y \notin FV(e)$ $(\lambda y.e_1)\{x := e\} = \lambda z.((e_1\{y := z\})\{x := e\})$ se $y \neq x$, $y \in FV(e)$ e z fresca $(\lambda x.e_1)\{x := e\} = \lambda x.e_1$
β-riduzione	$(\lambda x.e_1)e_2 \rightarrow e_1\{x := e_2\}$ $\frac{e_1 \rightarrow e'}{e_1e_2 \rightarrow e'e_2} \quad \frac{e_2 \rightarrow e'}{e_1e_2 \rightarrow e_1e'} \quad \frac{e \rightarrow e'}{\lambda x.e \rightarrow \lambda x.e'}$
Call-by-value	Call-by-name
$(\lambda x.e_1)v_2 \rightarrow e_1\{x := v_2\}$	$(\lambda x.e_1)e_2 \rightarrow e_1\{x := e_2\}$
$\frac{e_1 \rightarrow e'}{e_1e_2 \rightarrow e'e_2} \quad \frac{e_2 \rightarrow e'}{v_1e_2 \rightarrow v_1e'}$	$\frac{e_1 \rightarrow e'}{e_1e_2 \rightarrow e'e_2}$
β-equivalenza	$e_1 \equiv_\beta e_2$ se vale una tra: $e_1 \equiv_\alpha e_2 \quad e_1 \Rightarrow e_2 \quad e_2 \Rightarrow e_1 \quad e_1 \Rightarrow e \text{ e } e_2 \Rightarrow e$
Codifiche	$True = \lambda t.\lambda f.t \quad False = \lambda t.\lambda f.f$ $C_0 = \lambda z.\lambda s.z \quad C_1 = \lambda z.\lambda s.sz \quad C_n = \lambda z.\lambda s.s(...s(sz))$ $IF = \lambda c.\lambda then.\lambda else.c \text{ then } else$ $Plus = \lambda m.\lambda n.\lambda z.\lambda s.m(nzs)s$ $Times = \lambda m.\lambda n.mC_0(Plus\ n)$

Definizioni per le espressioni tipate

Sintassi $E ::= true \mid false \mid NV \mid if\ E\ then\ E\ else\ E$ $\mid succ\ E \mid pred\ E \mid isZero\ E$ $V ::= true \mid false \mid NV$ $NV ::= 0 \mid 1 \mid 2 \mid \dots$	Semantica (valutazione) $if\ true\ then\ E1\ else\ E2 \rightarrow E1$ $if\ false\ then\ E1\ else\ E2 \rightarrow E2$ $\frac{E \rightarrow E'}{if\ E\ then\ E1\ else\ E2 \rightarrow if\ E'\ then\ E1\ else\ E2}$ $\frac{E \rightarrow E'}{succ\ E \rightarrow succ\ E'} \quad \frac{m = n + 1}{succ\ n \rightarrow m}$ $\frac{E \rightarrow E'}{pred\ E \rightarrow pred\ E'} \quad \frac{n > 0 \quad m = n - 1}{pred\ n \rightarrow m}$ $pred\ 0 \rightarrow 0 \quad isZero\ 0 \rightarrow true$ $\frac{E \rightarrow E'}{isZero\ E \rightarrow isZero\ E'} \quad \frac{n > 0}{isZero\ n \rightarrow false}$
Tipi $T ::= Bool \mid Nat$	
Regole di controllo dei tipi $true : Bool \quad false : Bool \quad n : Nat$ $\frac{E : Nat}{succ\ E : Nat} \quad \frac{E : Nat}{pred\ E : Nat} \quad \frac{E : Nat}{isZero\ E : Bool}$ $\frac{E : Bool \quad E1 : T \quad E2 : T}{if\ E\ then\ E1\ else\ E2 : T}$	

Definizioni per il λ -calcolo tipato

Sintassi $V ::= true \mid false \mid fun\ x : \tau = e$ $e ::= x \mid fun\ x : \tau = e \mid Apply(e, e) \mid true \mid false$ $\mid if\ e\ then\ e\ else\ e$	Tipi $\tau ::= Bool \mid \tau \rightarrow \tau$
Regole di controllo dei tipi $\Gamma \vdash true : Bool$ $\Gamma \vdash false : Bool$ $\frac{\Gamma(x) = \tau}{\Gamma \vdash x : \tau}$ $\frac{\Gamma \vdash E : Bool \quad \Gamma \vdash E1 : \tau \quad \Gamma \vdash E2 : \tau}{\Gamma \vdash if\ E\ then\ E1\ else\ E2 : \tau}$ $\frac{\Gamma, x : \tau \vdash e : \tau_1 \rightarrow \tau_2}{\Gamma \vdash fun\ x : \tau_1 = e : \tau_1 \rightarrow \tau_2}$ $\frac{\Gamma \vdash e_1 : \tau_1 \rightarrow \tau_2 \quad \Gamma \vdash e_2 : \tau_1}{\Gamma \vdash Apply(e_1, e_2) : \tau_2}$	Semantica (valutazione) $if\ true\ then\ e_1\ else\ e_2 \rightarrow e_1$ $if\ false\ then\ e_1\ else\ e_2 \rightarrow e_2$ $\frac{e_1 \rightarrow e_4}{if\ e_1\ then\ e_2\ else\ e_3 \rightarrow if\ e_4\ then\ e_2\ else\ e_3}$ $Apply(fun\ x : \tau = e_1, v_2) \rightarrow e_1\{x := v_2\}$ $\frac{e_1 \rightarrow e'}{Apply(e_1, e_2) \rightarrow Apply(e', e_2)}$ $\frac{e_2 \rightarrow e'}{Apply(v_1, e_2) \rightarrow Apply(v_1, e')}$